

MyTalipapa

Public-Market Navigation & Stall-Management System

Data Flow Diagrams

Navigation Subsystem

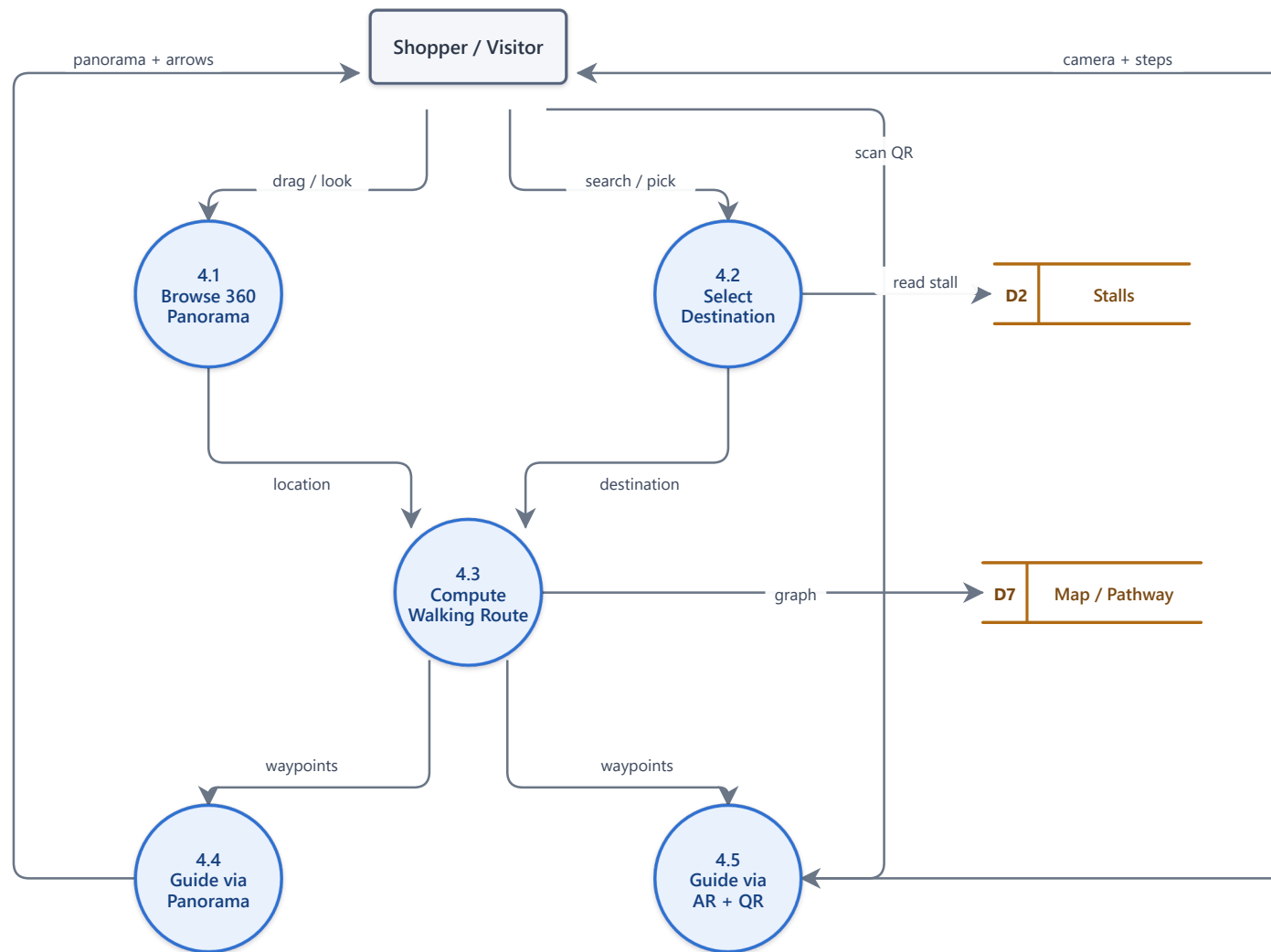
Yourdon–DeMarco notation

System Design Documentation · Version 1.0 · June 2026

DFD Level 2 — Process 4.0 Navigate & Find Stalls

Process 4.0 is decomposed into the wayfinding pipeline: the shopper browses the 360° panorama and selects a destination, the system computes a walking route, and the shopper is guided either through the panorama (arrows + mini-map) or the AR camera (with QR recognition and step detection).

- 4.1 Browse 360° Panorama
- 4.2 Select Destination
- 4.3 Compute Walking Route
- 4.4 Guide via Panorama
- 4.5 Guide via AR + QR



DFD Level 3 — Process 4.3 Compute Walking Route

Route computation is the pathfinding engine. It snaps the start and destination onto the nearest walkable aisles, builds an orthogonal aisle/corridor graph from the stored map geometry, runs Dijkstra's shortest-path algorithm, then simplifies the result into a minimal list of turn-by-turn waypoints — so routes follow real aisles and never cross a stall.

- 4.3.1 Snap Endpoints onto Aisles
- 4.3.2 Build Aisle / Corridor Graph

- 4.3.3 Run Dijkstra Shortest Path
- 4.3.4 Simplify to Turn-by-Turn Waypoints

